

```

const {
  Document, Packer, Paragraph, TextRun, Table, TableRow, TableCell,
  HeadingLevel, AlignmentType, LevelFormat, BorderStyle, WidthType,
  ShadingType, VerticalAlign, PageBreak, Header, Footer,
  TabStopType, TabStopPosition
} = require('docx');
const fs = require('fs');

const border = { style: BorderStyle.SINGLE, size: 1, color: "CCCCCC" };
const borders = { top: border, bottom: border, left: border, right: border };

function hCell(text, width, isHeader = false) {
  return new TableCell({
    borders,
    width: { size: width, type: WidthType.DXA },
    shading: isHeader ? { fill: "1E3A5F", type: ShadingType.CLEAR } : { fill: "F5F8FC", type: ShadingType.CLEAR },
    margins: { top: 100, bottom: 100, left: 140, right: 140 },
    verticalAlign: VerticalAlign.CENTER,
    children: [new Paragraph({
      alignment: AlignmentType.LEFT,
      children: [new TextRun({ text, font: "Arial", size: isHeader ? 20 : 20, bold: isHeader, color: isHeader ? "FFFFFF" : "222222" })]
    })]
  });
}

```

```

function dCell(text, width, shade = false) {
  return new TableCell({
    borders,
    width: { size: width, type: WidthType.DXA },
    shading: shade ? { fill: "EEF4FB", type: ShadingType.CLEAR } : { fill: "FFFFFF", type: ShadingType.CLEAR },
    margins: { top: 80, bottom: 80, left: 140, right: 140 },
    verticalAlign: VerticalAlign.CENTER,
    children: [new Paragraph({
      children: [new TextRun({ text, font: "Arial", size: 20, color: "333333" })]
    })]
  });
}

```

```

function heading1(text) {
  return new Paragraph({
    heading: HeadingLevel.HEADING_1,
    spacing: { before: 360, after: 160 },
    border: { bottom: { style: BorderStyle.SINGLE, size: 4, color: "1E3A5F", space: 1 } },
    children: [new TextRun({ text, font: "Arial", size: 36, bold: true, color: "1E3A5F" })]
  });
}

```

```

function heading2(text) {
  return new Paragraph({
    heading: HeadingLevel.HEADING_2,
    spacing: { before: 240, after: 120 },

```

```
children: [new TextRun({ text, font: "Arial", size: 28, bold: true, color: "2C5F8A" })]
});
}
```

```
function para(text, opts = {}) {
return new Paragraph({
spacing: { before: 80, after: 120 },
children: [new TextRun({ text, font: "Arial", size: 22, color: "333333", ...opts })]
});
}
```

```
function bullet(text) {
return new Paragraph({
numbering: { reference: "bullets", level: 0 },
spacing: { before: 60, after: 60 },
children: [new TextRun({ text, font: "Arial", size: 22, color: "333333" })]
});
}
```

```
function spacer() {
return new Paragraph({ spacing: { before: 80, after: 80 }, children: [new TextRun("")] });
}
```

```
const doc = new Document({
numbering: {
config: [
{
reference: "bullets",
levels: [{
level: 0, format: LevelFormat.BULLET, text: "\u2022", alignment: AlignmentType.LEFT,
style: { paragraph: { indent: { left: 720, hanging: 360 } } }
}]
}
],
},
styles: {
default: { document: { run: { font: "Arial", size: 22 } } },
paragraphStyles: [
{
id: "Heading1", name: "Heading 1", basedOn: "Normal", next: "Normal", quickFormat: true,
run: { size: 36, bold: true, font: "Arial", color: "1E3A5F" },
paragraph: { spacing: { before: 360, after: 160 }, outlineLevel: 0 }
},
{
id: "Heading2", name: "Heading 2", basedOn: "Normal", next: "Normal", quickFormat: true,
run: { size: 28, bold: true, font: "Arial", color: "2C5F8A" },
paragraph: { spacing: { before: 240, after: 120 }, outlineLevel: 1 }
}
]
},
sections: [{
properties: {
page: {
size: { width: 12240, height: 15840 },
```

```

margin: { top: 1440, right: 1440, bottom: 1440, left: 1440 }
},
headers: {
default: new Header({
children: [
new Paragraph({
border: { bottom: { style: BorderStyle.SINGLE, size: 4, color: "1E3A5F", space: 1 } },
spacing: { after: 100 },
children: [
new TextRun({ text: "RegLens AI ", font: "Arial", size: 20, bold: true, color: "1E3A5F" }),
new TextRun({ text: "\u2014 AI-Powered Regulatory Compliance Intelligence", font: "Arial", size:
20, color: "888888" })
]
})
]
})
},
footers: {
default: new Footer({
children: [
new Paragraph({
border: { top: { style: BorderStyle.SINGLE, size: 4, color: "1E3A5F", space: 1 } },
spacing: { before: 100 },
tabStops: [{ type: TabStopType.RIGHT, position: 9360 }],
children: [
new TextRun({ text: "github.com/zairahussain27/reglens-ai", font: "Arial", size: 18, color:
"888888" }),
new TextRun({ text: "\tPage ", font: "Arial", size: 18, color: "888888" }),
new PageNumber()
]
})
]
}),
children: [
// COVER
new Paragraph({
spacing: { before: 1440, after: 240 },
alignment: AlignmentType.CENTER,
children: [new TextRun({ text: "RegLens AI", font: "Arial", size: 72, bold: true, color: "1E3A5F" })]
}),
new Paragraph({
alignment: AlignmentType.CENTER,
spacing: { after: 120 },
children: [new TextRun({ text: "AI-Powered Regulatory Compliance Intelligence", font: "Arial", size:
32, color: "2C5F8A" })]
}),
new Paragraph({
alignment: AlignmentType.CENTER,
spacing: { after: 120 },
children: [new TextRun({ text: "For Indian FinTechs, Startups & MSMEs", font: "Arial", size: 26,
color: "555555", italics: true })]
}),

```

```
spacer(),
new Table({
width: { size: 7200, type: WidthType.DXA },
columnWidths: [2400, 4800],
rows: [
new TableRow({ children: [hCell("Category", 2400, true), hCell("Domain-Specialized AI with Compliance Guardrails", 4800, false)] }),
new TableRow({ children: [hCell("Target Users", 2400, true), hCell("FinTech Startups, NBFCs, MSMEs, E-Commerce", 4800, false)] }),
new TableRow({ children: [hCell("Regulatory Coverage", 2400, true), hCell("RBI, GST Council, SEBI, MCA, CBDT, FEMA", 4800, false)] }),
new TableRow({ children: [hCell("Core Technology", 2400, true), hCell("RAG + LLaMA 3.3 70B", 4800, false)] }),
new TableRow({ children: [hCell("Knowledge Base", 2400, true), hCell("11 official government documents, 1,356 indexed chunks", 4800, false)] }),
new TableRow({ children: [hCell("GitHub", 2400, true), hCell("github.com/zairahussain27/reglens-ai", 4800, false)] }),
]
}),
new Paragraph({ children: [new PageBreak()] }),
```

```
// 1. EXECUTIVE SUMMARY
heading1("1. Executive Summary"),
para("RegLens AI is an AI-powered regulatory compliance assistant designed to solve one of India's most persistent business challenges: the inability of FinTechs and MSMEs to accurately understand and operationalize complex government regulations."),
para("India has over 63 million MSMEs and 10,000+ FinTech startups, all operating under a dense web of regulations from RBI, GST Council, SEBI, MCA, and CBDT. These regulations are published as lengthy legal PDFs that are inaccessible to non-legal professionals. Professional compliance consulting costs Rs.25,000 to Rs.2,00,000 per year – unaffordable for most early-stage businesses."),
para("RegLens AI bridges this gap by combining Retrieval-Augmented Generation (RAG) with a fine-tuned reasoning prompt to deliver context-aware, source-cited compliance analysis in under 30 seconds. Unlike generic AI tools, RegLens AI never guesses – every answer is grounded in a real government document, and the system explicitly flags uncertainty rather than hallucinating."),
spacer(),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [4680, 4680],
rows: [
new TableRow({ children: [hCell("Metric", 4680, true), hCell("Value", 4680, true)] }),
new TableRow({ children: [dCell("Time to compliance analysis", 4680, false), dCell("< 30 seconds", 4680, false)] }),
new TableRow({ children: [dCell("Regulations covered", 4680, true), dCell("11 official government documents", 4680, true)] }),
new TableRow({ children: [dCell("Knowledge chunks indexed", 4680, false), dCell("1,356", 4680, false)] }),
new TableRow({ children: [dCell("Guardrails enforced", 4680, true), dCell("3 (scope, confidence, strict mode)", 4680, true)] }),
new TableRow({ children: [dCell("Hallucination prevention", 4680, false), dCell("RAG-grounded + temperature=0.1", 4680, false)] }),
new TableRow({ children: [dCell("Cost to user", 4680, true), dCell("Free (open source)", 4680, true)] }),
]
```

```
]
}),
new Paragraph({ children: [new PageBreak()] })),
```

```
// 2. PROBLEM STATEMENT
heading1("2. Problem Statement"),
para("Indian FinTech startups and MSMEs operate in a highly regulated environment governed by RBI, GST Council, SEBI, and MCA. Regulations are published as dense, legal-heavy PDFs that are difficult for non-legal professionals to interpret."),
heading2("Key Problems Identified"),
bullet("Unstructured Regulatory Data: Regulatory information is scattered across multiple government portals in unstructured PDF formats, making it difficult to extract and query relevant compliance requirements."),
bullet("No Business Context Mapping: Existing resources do not adapt regulations to specific business attributes such as industry type, revenue model, geography, or transaction nature."),
bullet("High Cost of Compliance Expertise: Professional legal and compliance consulting is expensive (Rs.25,000–Rs.2,00,000/year) and inaccessible to early-stage startups and MSMEs."),
bullet("Frequent Regulatory Updates: Indian financial regulations change frequently. Businesses struggle to track updates and understand how new circulars impact their current operations."),
bullet("Risk of Hallucination: Generic AI tools may provide incorrect, incomplete, or hallucinated legal interpretations – unacceptable in a strict compliance environment."),
bullet("No Explainable Decision Support: Existing tools rarely explain why a regulation applies or what exact action is required, reducing trust and making audits difficult."),
heading2("Impact of the Problem"),
bullet("Increased regulatory risk for FinTechs and MSMEs"),
bullet("Financial penalties and operational disruptions"),
bullet("Slower innovation due to fear of non-compliance"),
bullet("Unequal access to compliance knowledge – large enterprises with legal teams have a significant advantage"),
new Paragraph({ children: [new PageBreak()] })),
```

```
// 3. SOLUTION OVERVIEW
heading1("3. Solution Overview"),
para("RegLens AI is a system that reads real government regulations, understands your business, and tells you exactly which laws apply and what you must do – with explainable, auditable reasoning."),
heading2("What RegLens AI Does (Step by Step)"),
para("Step 1: Understands Your Business", { bold: true }),
para("The user inputs their business profile – type, industry, services, customer type, transaction nature, and revenue. The system converts this into a structured query for retrieval."),
para("Step 2: Reads Real Government Regulations", { bold: true }),
para("Instead of relying on training data, RegLens AI uses RAG to query a vectorized knowledge base of 11 official government documents – RBI circulars, GST rules, MSME guidelines, Companies Act, FEMA, and more."),
para("Step 3: Reasons About Applicability", { bold: true }),
para("This is the core innovation. The system does not just return documents – it reasons: does this law apply to THIS specific business? It outputs: Applicable / Conditionally Applicable / Not Applicable with explanation."),
para("Step 4: Generates Actionable Output", { bold: true }),
para("Instead of legal text, the user receives a structured compliance checklist, risk level (High/Medium/Low), specific action items, consequences of non-compliance, and citations to the exact source document."),
heading2("What Makes RegLens AI Different"),
bullet("Context-aware: Not generic – tailored to YOUR specific business profile"),
```

bullet("Grounded in real data: Uses actual government regulations, not AI memory"),  
bullet("Decision-focused: Not 'here is info' but 'here is what you must do'"),  
bullet("Strict mode: If unsure, it does NOT guess – it flags the uncertainty explicitly"),  
bullet("Source-cited: Every regulation answer cites the exact PDF it came from"),  
bullet("Auditable: Every decision is traceable to a specific government document"),  
new Paragraph({ children: [new PageBreak()] })),

// 4. SYSTEM ARCHITECTURE  
heading1("4. System Architecture"),  
para("RegLens AI is built as a multi-layer pipeline where each layer has a specific responsibility. The system is designed to fail safely – guardrails prevent the LLM from being called when confidence is low, eliminating hallucination risk."),  
spacer(),  
new Table({  
width: { size: 9360, type: WidthType.DXA },  
columnWidths: [2200, 2600, 4560],  
rows: [  
new TableRow({ children: [hCell("Layer", 2200, true), hCell("Components", 2600, true),  
hCell("Responsibility", 4560, true)] })),  
new TableRow({ children: [dCell("User Interface", 2200, false), dCell("Streamlit Web App", 2600, false), dCell("Business profile form, results display, disclaimer enforcement", 4560, false)] })),  
new TableRow({ children: [dCell("Guardrail Layer", 2200, true), dCell("Scope Validator + Confidence Assessor", 2600, true), dCell("Block out-of-scope requests and low-confidence retrievals before LLM call", 4560, true)] })),  
new TableRow({ children: [dCell("Retrieval Layer", 2200, false), dCell("Embedding Model + ChromaDB", 2600, false), dCell("Convert query to vector, search 1,356 chunks, return top 8 matches", 4560, false)] })),  
new TableRow({ children: [dCell("Reasoning Layer", 2200, true), dCell("LLaMA 3.3 70B + Prompt Template", 2600, true), dCell("Determine applicability, generate checklist, cite sources", 4560, true)] })),  
new TableRow({ children: [dCell("Output Layer", 2200, false), dCell("Structured Compliance Report", 2600, false), dCell("Display formatted report with risk level, actions, citations", 4560, false)] })),  
]  
}),  
new Paragraph({ children: [new PageBreak()] })),

// 5. ARCHITECTURE DIAGRAM (textual)  
heading1("5. Architecture Diagram"),  
para("The system comprises 5 layers with explicit fail paths at the guardrail stage."),  
spacer(),  
new Table({  
width: { size: 9360, type: WidthType.DXA },  
columnWidths: [9360],  
rows: [  
new TableRow({ children: [new TableCell({ borders, width: { size: 9360, type: WidthType.DXA },  
shading: { fill: "1E3A5F", type: ShadingType.CLEAR }, margins: { top: 100, bottom: 100, left: 180, right: 180 }, children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: "USER INTERFACE – Streamlit Web Application", font: "Arial", size: 22, bold: true, color: "FFFFFF" })] }))] })),  
new TableRow({ children: [new TableCell({ borders, width: { size: 9360, type: WidthType.DXA },  
shading: { fill: "2C5F8A", type: ShadingType.CLEAR }, margins: { top: 100, bottom: 100, left: 180, right: 180 }, children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: "GUARDRAIL LAYER – Scope Validator + Confidence Assessor (min 3 chunks)",

```
font: "Arial", size: 22, bold: true, color: "FFFFFF" }}] }}] })),
new TableRow({ children: [new TableCell({ borders, width: { size: 9360, type: WidthType.DXA },
shading: { fill: "EEF4FB", type: ShadingType.CLEAR }, margins: { top: 100, bottom: 100, left: 180,
right: 180 }, children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new
TextRun({ text: "RETRIEVAL LAYER — Embedding Model (all-MiniLM-L6-v2) + ChromaDB (1,356
chunks / 11 PDFs)", font: "Arial", size: 22, color: "1E3A5F" })] }}] }}] })),
new TableRow({ children: [new TableCell({ borders, width: { size: 9360, type: WidthType.DXA },
shading: { fill: "D5E8F0", type: ShadingType.CLEAR }, margins: { top: 100, bottom: 100, left: 180,
right: 180 }, children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new
TextRun({ text: "REASONING LAYER — LLaMA 3.3 70B via Groq API (temperature=0.1 — strict
factual mode)", font: "Arial", size: 22, color: "1E3A5F" })] }}] }}] })),
new TableRow({ children: [new TableCell({ borders, width: { size: 9360, type: WidthType.DXA },
shading: { fill: "EEF4FB", type: ShadingType.CLEAR }, margins: { top: 100, bottom: 100, left: 180,
right: 180 }, children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new
TextRun({ text: "OUTPUT LAYER — Compliance Report: citations + checklist + risk level", font:
"Arial", size: 22, color: "1E3A5F" })] }}] }}] })),
]
}),
para("Figure 1: RegLens AI System Architecture — showing all 5 layers, guardrail fail paths, and
the LLM reasoning layer.", { italics: true, color: "888888" }),
new Paragraph({ children: [new PageBreak()] })),
```

```
// 6. AGENT COMMUNICATION FLOW
heading1("6. Agent Communication Flow"),
para("Each step passes only on success. Steps 2 and 5 have explicit FAIL exits that return safe rejection messages without calling the LLM."),
spacer(),
...[
"1. User Input",
"2. Scope Check ✖ FAIL: Out of scope ✖ reject",
"3. Query Builder",
"4. Vector Search (ChromaDB)",
"5. Confidence Check ✖ FAIL: Low confidence ✖ reject",
"6. Prompt Assembly — Profile + Chunks + Template",
"7. LLaMA 3.3 70B (Groq) — temperature=0.1",
"8. Structured Report Generated",
"9. Streamlit UI Display"
].map(step => bullet(step)),
new Paragraph({ children: [new PageBreak()] }),
```

```
// 7. COMPONENT BREAKDOWN
heading1("7. Component Breakdown"),
heading2("7.1 Ingestion Pipeline (src/ingest.py)",
para("The ingestion pipeline is responsible for processing all regulatory PDFs and storing them in the vector database. It runs once during setup and creates the knowledge base."),
bullet("Reads all PDFs from the regulations/ folder using pdfplumber"),
bullet("Splits each document into 500-word chunks with 50-word overlap to preserve context across boundaries"),
bullet("Generates dense vector embeddings for each chunk using sentence-transformers (all-MiniLM-L6-v2)"),
bullet("Stores chunks, embeddings, and metadata (source filename) in ChromaDB persistent vector store"),
bullet("Total output: 1,356 chunks indexed across 11 government documents"),
heading2("7.2 Retrieval Engine (src/retriever.py"),
```

```
para("The retrieval engine converts a business profile query into a vector and searches the
knowledge base for the most semantically relevant regulation chunks."),
bullet("Accepts a natural language query describing the business"),
bullet("Generates a query embedding using the same model used during ingestion"),
bullet("Performs cosine similarity search in ChromaDB"),
bullet("Returns top 8 most relevant chunks along with their source filenames"),
bullet("Source filenames are passed to the LLM for citation in the output"),
heading2("7.3 Compliance Engine (src/compliance_engine.py)"),
para("The compliance engine is the core of RegLens AI. It orchestrates the full pipeline: scope
validation, retrieval, confidence assessment, prompt assembly, and LLM reasoning."),
bullet("Scope Validator checks if the business industry is in the supported domain list"),
bullet("Confidence Assessor verifies at least 3 relevant chunks were retrieved"),
bullet("Prompt template is loaded from prompts/compliance.txt"),
bullet("Business profile and retrieved chunks are injected into the prompt"),
bullet("LLaMA 3.3 70B is called via Groq API with temperature=0.1 for strict factual output"),
bullet("System prompt enforces: never guess, always cite source, flag uncertainty explicitly"),
heading2("7.4 Streamlit UI (src/app.py)"),
para("The user interface is built with Streamlit for rapid deployment and clean presentation."),
bullet("Two-column layout: business details on left, customer/transaction details on right"),
bullet("Six input fields: Business Type, Industry, Services, Customer Type, Transaction Type,
Revenue"),
bullet("Single 'Run Compliance Check' button triggers the full pipeline"),
bullet("Results rendered as formatted markdown with emoji indicators"),
bullet("Sidebar shows regulatory coverage and system description"),
bullet("Footer disclaimer enforced on every result page"),
new Paragraph({ children: [new PageBreak()] })),
```

// 8. KNOWLEDGE BASE

```
heading1("8. Knowledge Base & Regulatory Coverage"),
para("RegLens AI uses 11 official government documents as its sole source of truth. No
information is sourced from the model's training data for legal facts."),
spacer(),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [400, 3200, 1560, 2800, 1400],
rows: [
new TableRow({ children: [hCell("#", 400, true), hCell("Document", 3200, true), hCell("Authority",
1560, true), hCell("Coverage", 2800, true), hCell("Chunks", 1400, true)] }),
...[
["1", "RBI KYC Master Direction", "RBI", "Identity verification, customer due diligence", "76"],
["2", "RBI Payment Aggregators Guidelines", "RBI", "Payment gateway, aggregator compliance",
"15"],
["3", "RBI PPI Master Directions", "RBI", "Prepaid payment instruments, wallets", "34"],
["4", "RBI Digital Lending Guidelines 2022", "RBI", "Digital lending, BNPL, LSP framework", "8"],
["5", "RBI NBFC Master Direction", "RBI", "NBFC registration, operations, reporting", "4"],
["6", "RBI Fair Practices Code", "RBI", "NBFC customer treatment, transparency", "347"],
["7", "CGST Rules 2017", "GST Council", "GST registration, returns, composition scheme", "178"],
["8", "MSME Udyam Registration", "Ministry of MSME", "MSME definition, registration process",
"18"],
["9", "Income Tax TDS Section 194", "CBDT", "TDS provisions for business payments", "607"],
["10", "FEMA Basic Compliance", "RBI", "Foreign exchange, cross-border transactions", "41"],
["11", "Companies Act 2013", "MCA", "Company registration, annual filing, directors", "28"],
].map((row, i) => new TableRow({ children: row.map((cell, j) => dCell(cell, [400, 3200, 1560, 2800,
1400][j], i % 2 === 0)) })),
```



```
new TableRow({ children: [hCell("TOTAL", 400, true), hCell("", 3200, true), hCell("", 1560, true),
hCell("", 2800, true), hCell("1,356", 1400, true)] })),
],
}),
new Paragraph({ children: [new PageBreak()] })),

// 9. GUARDRAIL & SAFETY
heading1("9. Guardrail & Safety System"),
para("RegLens AI implements three enforced guardrails that together prevent hallucination, out-
of-scope responses, and overconfident legal claims."),
heading2("Guardrail 1: Scope Validator"),
para("Checks if the business industry field matches any supported domain keyword before any
processing begins."),
para("Trigger: Industry input does not match: fintech, digital payments, lending, nbfc, msme,
manufacturing, gst, ecommerce"),
para("Response: Returns 'Out of Scope' message. No embedding, no retrieval, no LLM call."),
heading2("Guardrail 2: Confidence Assessor"),
para("After retrieval, checks if enough relevant regulation chunks were found to support a
confident answer."),
para("Trigger: Fewer than 3 chunks retrieved from ChromaDB for the given query"),
para("Response: Returns 'Insufficient Data' message. No LLM call made."),
heading2("Guardrail 3: Strict Mode Prompt"),
para("The system prompt instructs the LLM to flag uncertainty rather than guess, and to return
'INSUFFICIENT DATA' if the context does not support a confident answer."),
para("Trigger: LLM cannot find grounding in retrieved context for a specific regulation"),
para("Response: LLM outputs INSUFFICIENT DATA flag for that regulation, not a hallucinated
answer."),
spacer(),
heading2("Error Handling"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [3120, 4320, 1920],
rows: [
new TableRow({ children: [hCell("Scenario", 3120, true), hCell("System Response", 4320, true),
hCell("LLM Called?", 1920, true)] })),
new TableRow({ children: [dCell("Business outside supported domain", 3120), dCell("Out of Scope
rejection message", 4320), dCell("No", 1920)] })),
new TableRow({ children: [dCell("Fewer than 3 chunks retrieved", 3120, true), dCell("Insufficient
Data warning", 4320, true), dCell("No", 1920, true)] })),
new TableRow({ children: [dCell("LLM uncertain about a regulation", 3120), dCell("UNCERTAIN
flag in output", 4320), dCell("Yes", 1920)] })),
new TableRow({ children: [dCell("Regulation not in retrieved context", 3120, true),
dCell("INSUFFICIENT DATA for that item", 4320, true), dCell("Yes", 1920, true)] })),
new TableRow({ children: [dCell("API call failure", 3120), dCell("Streamlit error message
displayed", 4320), dCell("Attempted", 1920)] })),
],
}),
new Paragraph({ children: [new PageBreak()] })),

// 10. TECH STACK
heading1("10. Tech Stack"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [2000, 3160, 4200],
```

```

rows: [
  new TableRow({ children: [hCell("Layer", 2000, true), hCell("Technology", 3160, true),
    hCell("Purpose", 4200, true)] }),
  new TableRow({ children: [dCell("Frontend", 2000), dCell("Streamlit", 3160), dCell("Web UI –
    business profile form + results display", 4200)] }),
  new TableRow({ children: [dCell("LLM", 2000, true), dCell("LLaMA 3.3 70B via Groq API", 3160,
    true), dCell("Compliance reasoning, applicability decision, checklist generation", 4200, true)] }),
  new TableRow({ children: [dCell("Vector DB", 2000), dCell("ChromaDB (persistent)", 3160),
    dCell("Store and search 1,356 regulation chunks by semantic similarity", 4200)] }),
  new TableRow({ children: [dCell("Embeddings", 2000, true), dCell("all-MiniLM-L6-v2 (sentence-
    transformers)", 3160, true), dCell("Convert text to dense vectors for similarity search", 4200,
    true)] }),
  new TableRow({ children: [dCell("PDF Parsing", 2000), dCell("pdfplumber", 3160), dCell("Extract
    text from government PDF documents", 4200)] }),
  new TableRow({ children: [dCell("Backend", 2000, true), dCell("Python 3.12", 3160, true),
    dCell("Pipeline orchestration, API calls, guardrail logic", 4200, true)] }),
  new TableRow({ children: [dCell("Environment", 2000), dCell("python-dotenv", 3160), dCell("Secure
    API key management via .env file", 4200)] }),
  new TableRow({ children: [dCell("Version Control", 2000, true), dCell("GitHub", 3160, true),
    dCell("Source code hosting and collaboration", 4200, true)] }),
]
}),
new Paragraph({ children: [new PageBreak()] }),

```

```

// 11. SETUP & INSTALLATION
heading1("11. Setup & Installation"),
heading2("Prerequisites"),
bullet("Python 3.10 or higher"),
bullet("Git"),
bullet("Groq API key (free at console.groq.com)"),
heading2("Step 1: Clone the Repository"),
new Paragraph({
  spacing: { before: 80, after: 80 },
  shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
  children: [new TextRun({ text: "git clone https://github.com/zairahussain27/reglens-ai", font:
    "Courier New", size: 20, color: "1E3A5F" })]
}),
new Paragraph({
  spacing: { before: 0, after: 80 },
  shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
  children: [new TextRun({ text: "cd reglens-ai", font: "Courier New", size: 20, color: "1E3A5F" })]
}),
heading2("Step 2: Install Dependencies"),
new Paragraph({
  spacing: { before: 80, after: 80 },
  shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
  children: [new TextRun({ text: "pip install -r requirements.txt", font: "Courier New", size: 20, color:
    "1E3A5F" })]
}),
heading2("Step 3: Set Up API Key"),
para("Create a .env file in the root folder and add your Groq API key:"),
new Paragraph({
  spacing: { before: 80, after: 80 },
  shading: { fill: "F0F4F8", type: ShadingType.CLEAR },

```

```

children: [new TextRun({ text: "GROQ_API_KEY=your_groq_api_key_here", font: "Courier New",
size: 20, color: "1E3A5F" })]
}),
heading2("Step 4: Ingest Regulatory Documents"),
para("Run the ingestion script to process all PDFs and build the vector database:"),
new Paragraph({
spacing: { before: 80, after: 80 },
shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
children: [new TextRun({ text: "python src/ingest.py", font: "Courier New", size: 20, color:
"1E3A5F" })]
}),
para("This will process all 11 PDFs in the regulations/ folder and create a chroma_db/ directory.
This step takes 2–5 minutes and only needs to be run once."),
heading2("Step 5: Launch the Application"),
new Paragraph({
spacing: { before: 80, after: 80 },
shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
children: [new TextRun({ text: "streamlit run src/app.py", font: "Courier New", size: 20, color:
"1E3A5F" })]
}),
para("The application will open automatically at http://localhost:8501"),
heading2("requirements.txt Contents"),
...["streamlit", "pdfplumber", "chromadb", "sentence-transformers", "groq", "python-dotenv",
"fastapi", "uvicorn"].map(pkg =>
new Paragraph({
spacing: { before: 40, after: 40 },
shading: { fill: "F0F4F8", type: ShadingType.CLEAR },
children: [new TextRun({ text: pkg, font: "Courier New", size: 20, color: "1E3A5F" })]
})
),
new Paragraph({ children: [new PageBreak()] }),

```

```
// 12. HOW TO USE
```

```

heading1("12. How to Use"),
para("Once the application is running, follow these steps to get a compliance analysis:"),
bullet("Step 1: Open the Application — Navigate to http://localhost:8501 in your browser."),
bullet("Step 2: Fill Business Type — Select your legal entity type: Private Limited Company, LLP,
Sole Proprietorship, Partnership Firm, or OPC."),
bullet("Step 3: Select Industry — Choose your primary industry: FinTech Digital Payments,
FinTech Lending/NBFC, MSME Manufacturing, MSME Services, E-Commerce, or SaaS/
Technology."),
bullet("Step 4: Describe Your Services — Type a brief description of what your business does."),
bullet("Step 5: Select Customer & Transaction Type — Specify who your customers are and your
primary transaction type."),
bullet("Step 6: Select Annual Revenue — Choose your revenue band. This affects which
compliance thresholds apply."),
bullet("Step 7: Run Compliance Check — Click the 'Run Compliance Check' button. The analysis
takes 15–30 seconds."),
bullet("Step 8: Review the Report — The compliance report will appear below with applicable
regulations, risk level, checklist, and recommended next steps."),
new Paragraph({ children: [new PageBreak()] }),

```

```
// 13. SAMPLE OUTPUT
```

```

heading1("13. Sample Output"),

```

```

heading2("Input Profile"),
para("Business Type: Private Limited Company | Industry: FinTech - Digital Payments | Services: Online payment gateway, wallet services, UPI transactions | Revenue: Under Rs.1 Crore"),
heading2("Sample Output Structure"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [3000, 6360],
rows: [
new TableRow({ children: [hCell("Section", 3000, true), hCell("Content", 6360, true)] }),
new TableRow({ children: [dCell("Business Profile Summary", 3000), dCell("2-line AI-generated summary of the business based on inputs", 6360)] }),
new TableRow({ children: [dCell("Applicable Regulations", 3000, true), dCell("List of regulations with status and source citation", 6360, true)] }),
new TableRow({ children: [dCell("Overall Risk Level", 3000), dCell("Red / Yellow / Green risk indicator with one-line explanation", 6360)] }),
new TableRow({ children: [dCell("Flags & Uncertainties", 3000, true), dCell("Any items the system was not confident about – never hidden", 6360, true)] }),
new TableRow({ children: [dCell("Recommended Next Steps", 3000), dCell("3 prioritized actions the business should take immediately", 6360)] }),
]
}),
spacer(),
heading2("Regulation Status Indicators"),
bullet("Applicable: This regulation clearly applies to your business based on retrieved context"),
bullet("Conditionally Applicable: This regulation may apply depending on specific activities you engage in"),
bullet("Not Applicable: This regulation does not apply to your described business model"),
bullet("INSUFFICIENT DATA: The system could not find enough context to make a confident determination"),
new Paragraph({ children: [new PageBreak()] }),

```

```

// 14. SCALABILITY & ROADMAP

```

```

heading1("14. Scalability & Roadmap"),
heading2("Immediate Scalability (No Code Changes Required)"),
bullet("Add more PDFs to regulations/ folder and re-run ingest.py to expand coverage"),
bullet("Add new industries to the scope validator list in compliance_engine.py"),
bullet("Swap Groq for any OpenAI-compatible API endpoint"),
heading2("Short-term Roadmap (1–3 months)"),
bullet("Multilingual support – add Hindi and regional language interfaces using multilingual embedding models"),
bullet("Real-time regulatory updates – schedule automated scraping of RBI/GST portals for new circulars"),
bullet("WhatsApp/Telegram bot interface for mobile-first MSME users"),
bullet("Expand to 50+ government documents for deeper coverage"),
heading2("Long-term Roadmap (6–12 months)"),
bullet("Production deployment on AWS/GCP with Pinecone vector DB for scale"),
bullet("SEBI regulations for investment and securities compliance"),
bullet("Auto-filing assistance – generate draft compliance filings from analysis"),
bullet("Compliance history tracking – alert businesses when regulations affecting them are updated"),
bullet("API product – white-label RegLens AI for banks and compliance platforms"),
new Paragraph({ children: [new PageBreak()] }),

```

```

// 15. IMPACT & BUSINESS VALUE

```

```

heading1("15. Impact & Business Value"),
heading2("15.1 Time Savings"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [3000, 2000, 2000, 2360],
rows: [
new TableRow({ children: [hCell("Task", 3000, true), hCell("Manual Process", 2000, true),
hCell("With RegLens AI", 2000, true), hCell("Time Saved", 2360, true)] }),
new TableRow({ children: [dCell("Identify applicable regulations", 3000), dCell("4–8 hours", 2000),
dCell("Automatic", 2000), dCell("4–8 hours", 2360)] }),
new TableRow({ children: [dCell("Read & interpret legal PDFs", 3000, true), dCell("6–12 hours",
2000, true), dCell("Automatic", 2000, true), dCell("6–12 hours", 2360, true)] }),
new TableRow({ children: [dCell("Consult CA / legal advisor", 3000), dCell("2–5 hours + wait",
2000), dCell("Not needed", 2000), dCell("2–5 hours", 2360)] }),
new TableRow({ children: [dCell("Total per compliance review", 3000, true), dCell("12–25 hours",
2000, true), dCell("~2 hours", 2000, true), dCell("10–23 hours", 2360, true)] }),
new TableRow({ children: [dCell("Annual (2 reviews/year)", 3000), dCell("24–50 hours", 2000),
dCell("~4 hours", 2000), dCell("20–46 hours", 2360)] }),
]
}),
spacer(),
heading2("15.2 Cost Savings Per Business"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [3000, 2200, 2000, 2160],
rows: [
new TableRow({ children: [hCell("Cost Item", 3000, true), hCell("Current Cost", 2200, true),
hCell("With RegLens AI", 2000, true), hCell("Annual Saving", 2160, true)] }),
new TableRow({ children: [dCell("Compliance consultant", 3000), dCell("Rs.50,000/year", 2200),
dCell("Rs.0", 2000), dCell("Rs.50,000", 2160)] }),
new TableRow({ children: [dCell("CA for GST + TDS advice", 3000, true), dCell("Rs.25,000/year",
2200, true), dCell("Rs.0", 2000, true), dCell("Rs.25,000", 2160, true)] }),
new TableRow({ children: [dCell("GST penalty avoidance", 3000), dCell("Rs.15,000/incident",
2200), dCell("Rs.0 if compliant", 2000), dCell("Rs.15,000", 2160)] }),
new TableRow({ children: [hCell("Total saving per MSME/year", 3000, true), hCell("", 2200, true),
hCell("", 2000, true), hCell("Rs.75,000 – Rs.90,000", 2160, true)] }),
]
}),
spacer(),
heading2("15.3 Aggregate Impact at Scale"),
new Table({
width: { size: 9360, type: WidthType.DXA },
columnWidths: [2200, 1800, 2160, 1800, 1400],
rows: [
new TableRow({ children: [hCell("Scenario", 2200, true), hCell("Adoption Rate", 1800, true),
hCell("MSMEs Reached", 2160, true), hCell("Cost Saved/Year", 1800, true), hCell("Hours Saved",
1400, true)] }),
new TableRow({ children: [dCell("Conservative", 2200), dCell("0.1%", 1800), dCell("63,000", 2160),
dCell("Rs.472.5 crore", 1800), dCell("1.89M", 1400)] }),
new TableRow({ children: [dCell("Moderate", 2200, true), dCell("1%", 1800, true), dCell("6,30,000",
2160, true), dCell("Rs.4,725 crore", 1800, true), dCell("18.9M", 1400, true)] }),
new TableRow({ children: [dCell("Optimistic", 2200), dCell("5%", 1800), dCell("31,50,000", 2160),
dCell("Rs.23,625 crore", 1800), dCell("94.5M", 1400)] }),
]
}

```

```

    }),
    spacer(),
    heading2("15.4 Non-Financial Impact"),
    bullet("Democratization: Small businesses get the same compliance quality as large enterprises with dedicated legal teams."),
    bullet("Innovation Unlock: Startups can enter regulated markets faster without fear of unknown compliance requirements."),
    bullet("Trust & Transparency: Source-cited answers build regulatory trust – businesses know exactly which law applies and why."),
    bullet("Financial Formalization: More MSMEs entering formal compliance systems contributes to GST revenue and financial inclusion."),
    bullet("Reduced Inequality: Levels the playing field between large enterprises and small businesses in navigating regulatory complexity."),
    spacer(),
    new Paragraph({
    alignment: AlignmentType.CENTER,
    spacing: { before: 360, after: 120 },
    border: { top: { style: BorderStyle.SINGLE, size: 4, color: "1E3A5F", space: 1 } },
    children: [new TextRun({ text: "RegLens AI – AI-Powered Regulatory Compliance for Indian FinTechs & MSMEs", font: "Arial", size: 22, bold: true, color: "1E3A5F" })]
    }),
    new Paragraph({
    alignment: AlignmentType.CENTER,
    children: [new TextRun({ text: "GitHub: https://github.com/zairahussain27/reglens-ai", font: "Arial", size: 20, color: "2C5F8A" })]
    }),
    ]
    })
    });

Packer.toBuffer(doc).then(buffer => {
fs.writeFileSync("/mnt/user-data/outputs/RegLens_AI_Documentation.docx", buffer);
console.log("Done");
});

```